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(KEPINTARAN BUATAN)
SECTION 03

PROJECT PROPOSAL:
PARKING DETECTION APPLICATION

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Problem Statement

In many urban and campus environments, drivers often experience difficulty finding available parking spaces, particularly during peak hours such as early mornings when most people go to work and late evenings when most people just finished their daily shifts. The lack of real-time information about parking availability forces drivers to circle around parking areas aimlessly, resulting in wasted time, increased fuel consumption, and added traffic congestion. This not only causes frustration among drivers but also contributes to higher carbon emissions and inefficient use of parking infrastructure.

Traditional parking systems, which rely on manual monitoring or simple sensor-based approaches, are limited in accuracy and scalability. Therefore, there is a pressing need for an AI-powered parking detection system that can automatically identify vacant spaces through real-time image analysis and provide up-to-date parking availability to users. Such a system would improve driver convenience, reduce traffic congestion, and enhance the overall efficiency of urban mobility.

Process of Empathize in Design Thinking



Process of Define in Design Thinking

USER	NEED	INSIGHT
Drivers who frequently travel in busy city or town areas.	• Need to find available parking spaces quickly without spending excessive time searching. • Need accurate and real-time information about parking availability to avoid circling multiple times. • Need a convenient navigation feature that can guide them directly to the nearest empty parking spot. • Need a reliable system that minimizes frustration and reduces the chances of parking fines or arriving late.	• Many drivers waste their time daily looking for parking, especially during peak hours, leading to stress and late arrivals. • The lack of live parking updates can cause drivers to circle parking areas repeatedly, worsening traffic congestion and fuel usage. • Drivers feel frustrated when parking apps or manual systems are inaccurate or outdated. • An AI-powered parking detection system that provides real-time updates and navigation support can significantly improve user satisfaction, reduce stress and save time.

The Goal of AI solution

1. **Efficient Parking Detection:**
To accurately detect and monitor available parking spaces using AI-powered image recognition in real time.
2. **Reduced Search Time:**
Shorten the time drivers spend searching for a parking spot, by guiding them directly to available spots, improving overall parking efficiency.
3. **Decreased Traffic Congestion:**
Minimize unnecessary vehicle movement caused by drivers circling nonstop for parking, which can aid in reducing traffic density and emissions in urban areas.
4. **Enhanced User Experience:**
Provide users with an easy to use application that delivers live parking availability and directions, promoting convenience and reliability.
5. **Data Driven Urban Planning:**
Contribute to local authorities with data analytics on parking usage patterns and periods to support better infrastructure design and city management

AI Solution

To address the common challenge of finding available parking spaces in congested urban areas, our project proposes an AI Smart Parking Detection System. This solution aims to help drivers locate vacant parking spaces quickly and efficiently, reducing time wastage, traffic congestion, and fuel consumption.

The system utilizes computer vision technology powered by Artificial Intelligence (AI) and Machine Learning (ML) algorithms. Cameras are strategically installed within parking areas to capture real-time images of parking lots. The AI model processes these images to detect whether each parking spot is occupied or vacant by identifying vehicles and analyzing space patterns. The processed data is then sent to a cloud database, where it is updated in real time.

Drivers can access this information through a mobile or web application, which displays the current availability of parking spaces and guides users to the nearest vacant spot. The system can also provide parking statistics and trends to authorities, helping them optimize parking infrastructure and improve management efficiency.

Ultimately, this AI solution enhances urban convenience and mobility by minimizing the frustration and inefficiency associated with searching for parking.